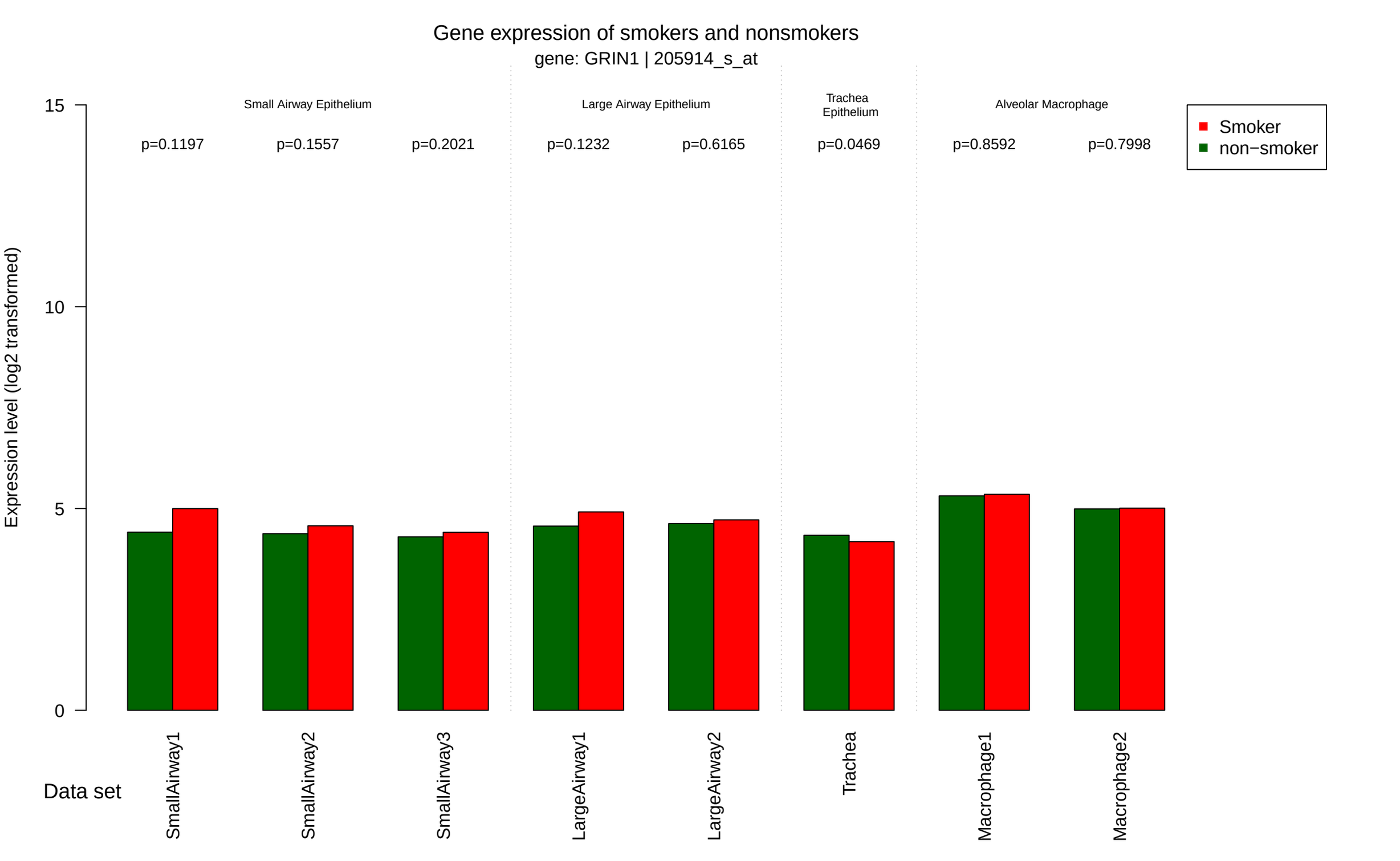
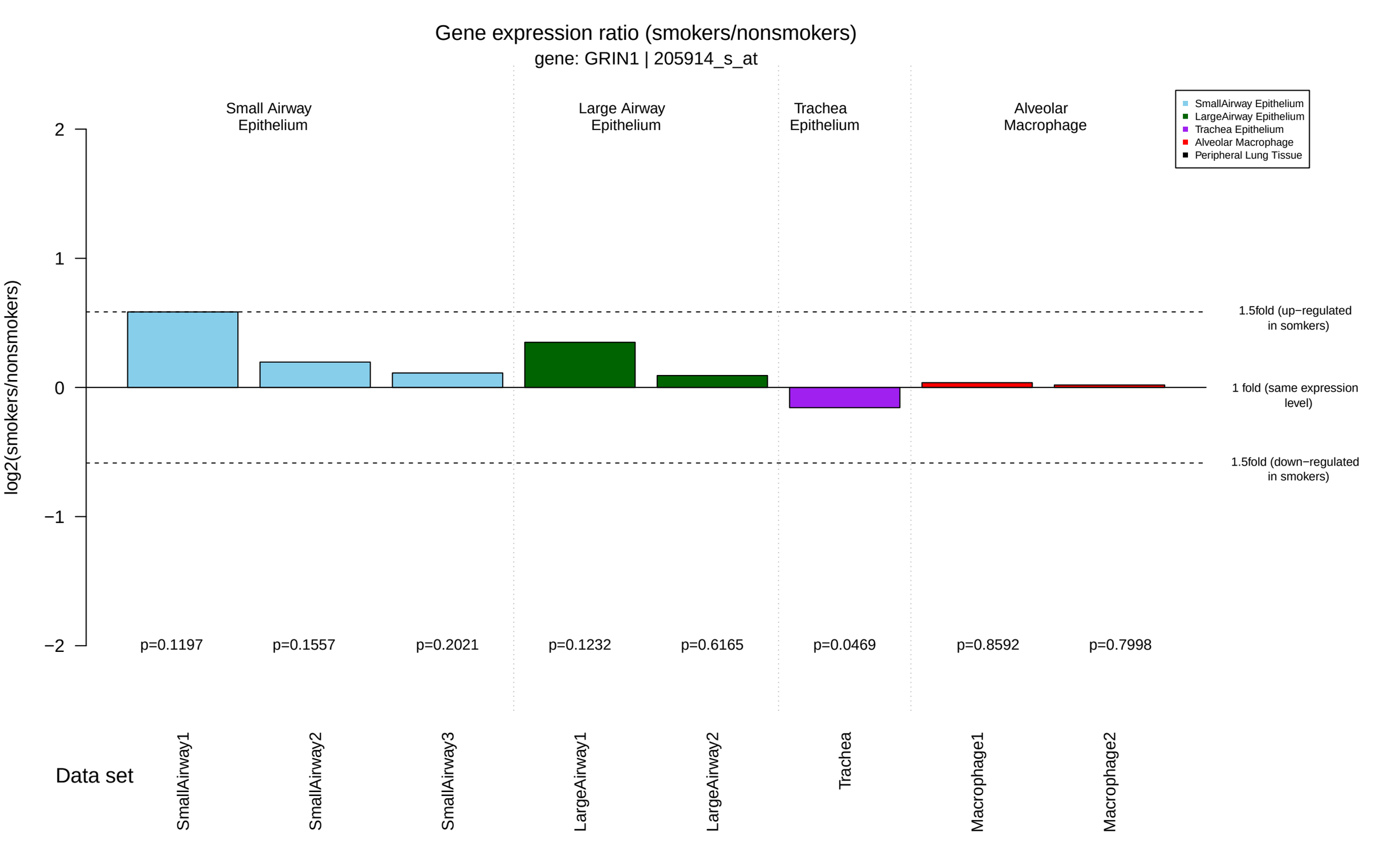
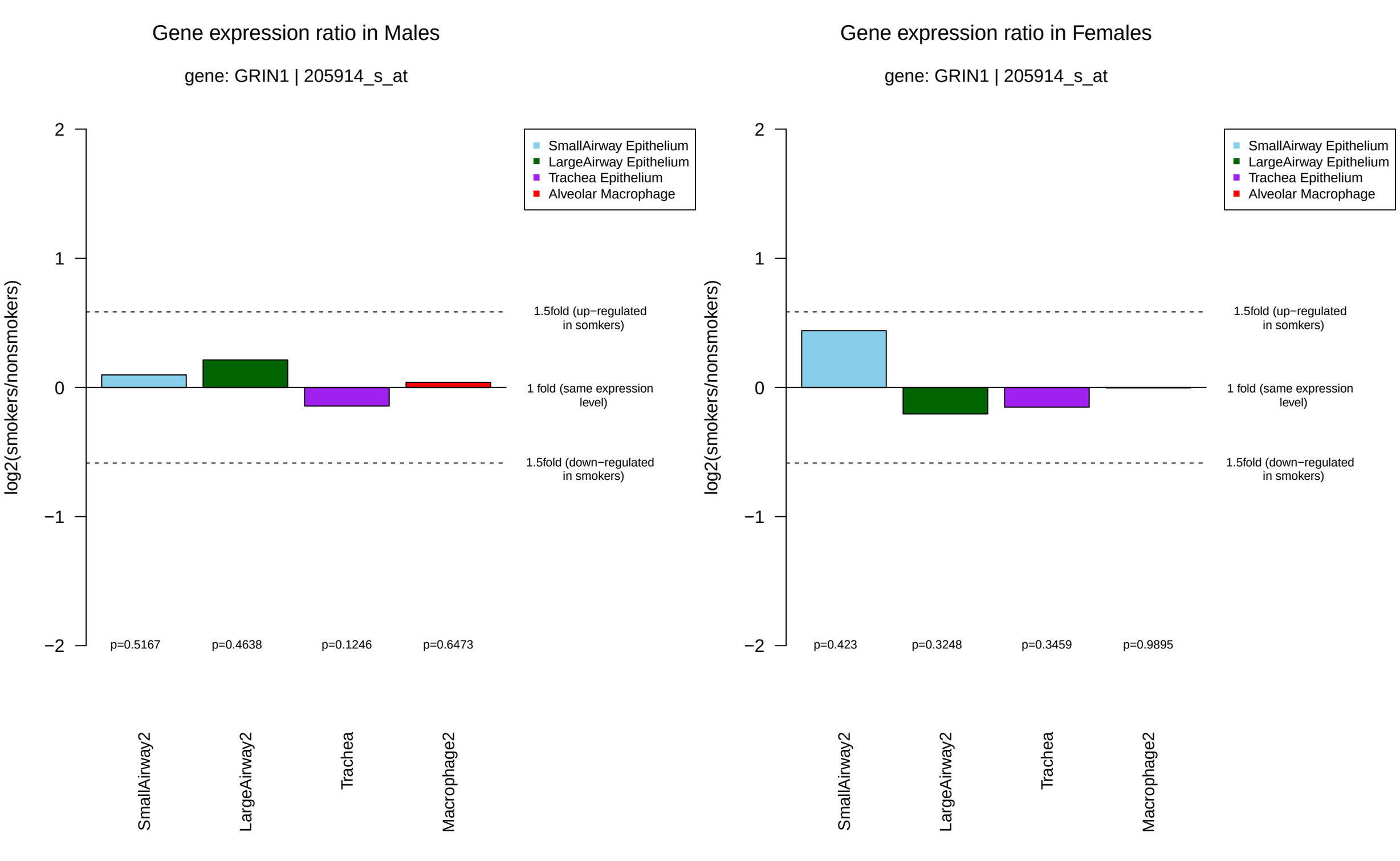
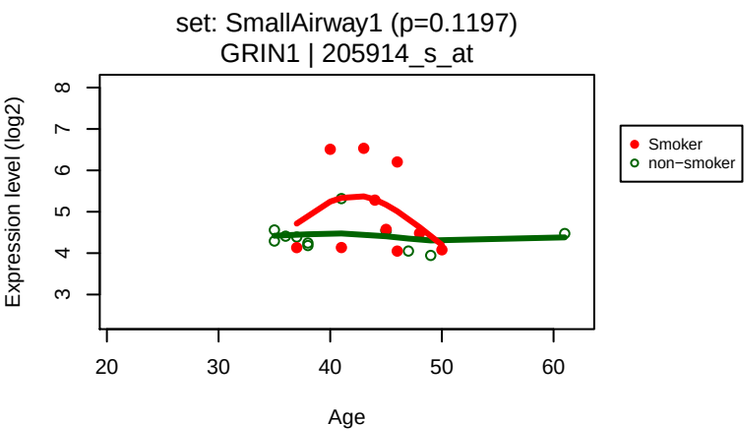
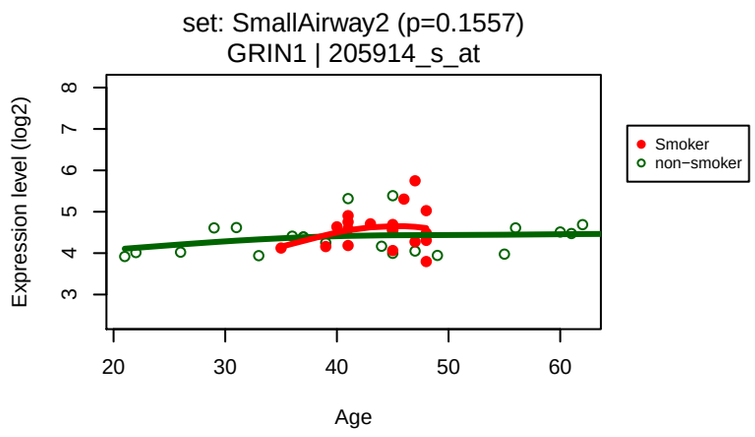
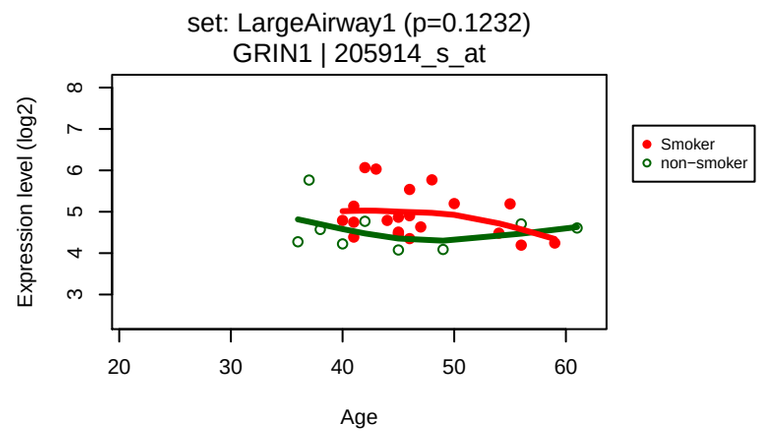

Gene expression of smokers and nonsmokers
gene: GRIN1 | 205914_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1197
p=0.1557
p=0.2021
p=0.1232
p=0.6165
p=0.0469
p=0.8592
p=0.7998
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

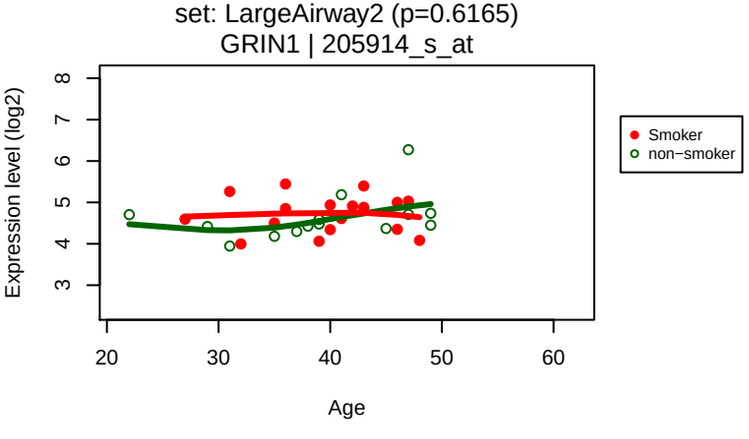

Gene expression ratio (smokers/nonsmokers)
gene: GRIN1 | 205914_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1197
p=0.1557
p=0.2021
p=0.1232
p=0.6165
p=0.0469
p=0.8592
p=0.7998
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

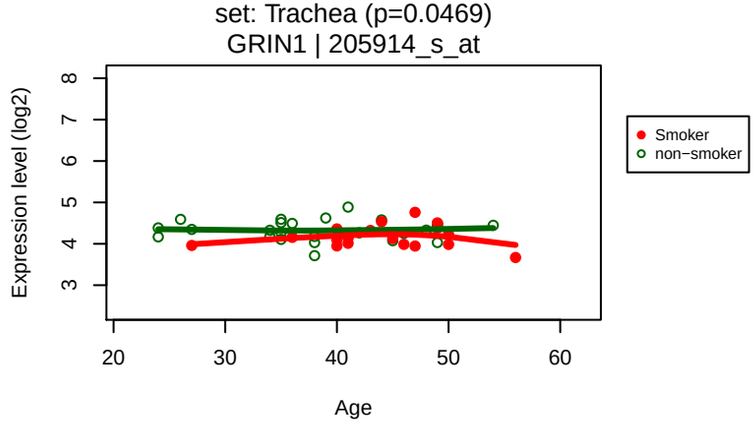

Gene expression ratio in Males
gene: GRIN1 | 205914_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5167
p=0.4638
p=0.1246
p=0.6473
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIN1 | 205914_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.423
p=0.3248
p=0.3459
p=0.9895
SmallAirway2
LargeAirway2
Trachea
Macrophage2

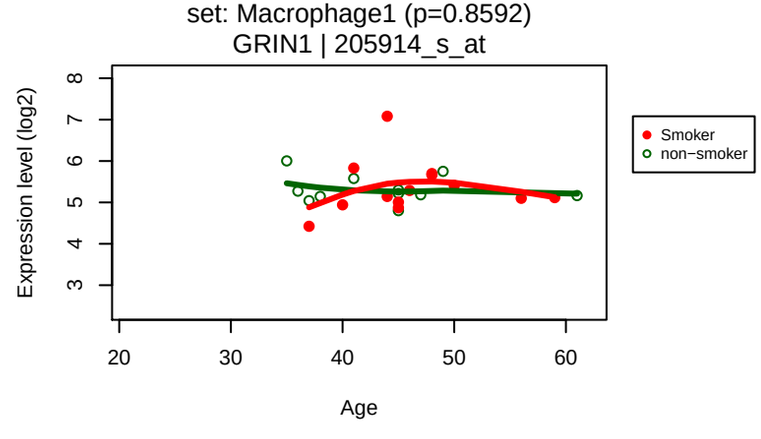

set: SmallAirway1 (p=0.1197)
GRIN1 | 205914_s_at
Expression level (log2)
Age
Smoker
non−smoker

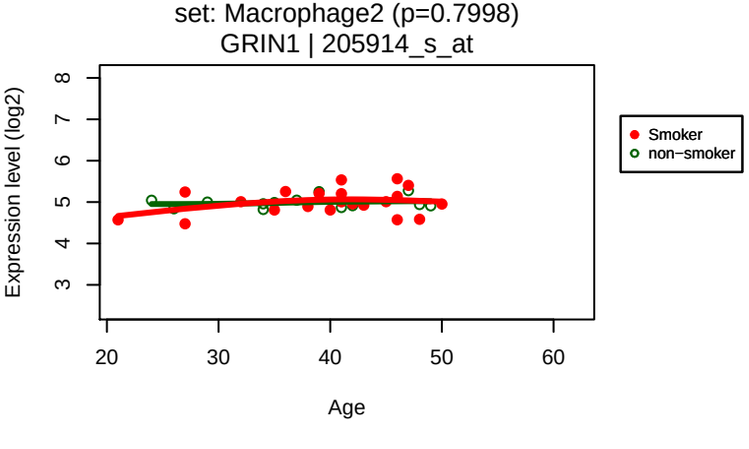

set: SmallAirway2 (p=0.1557)
GRIN1 | 205914_s_at
Expression level (log2)
Age
Smoker
non−smoker

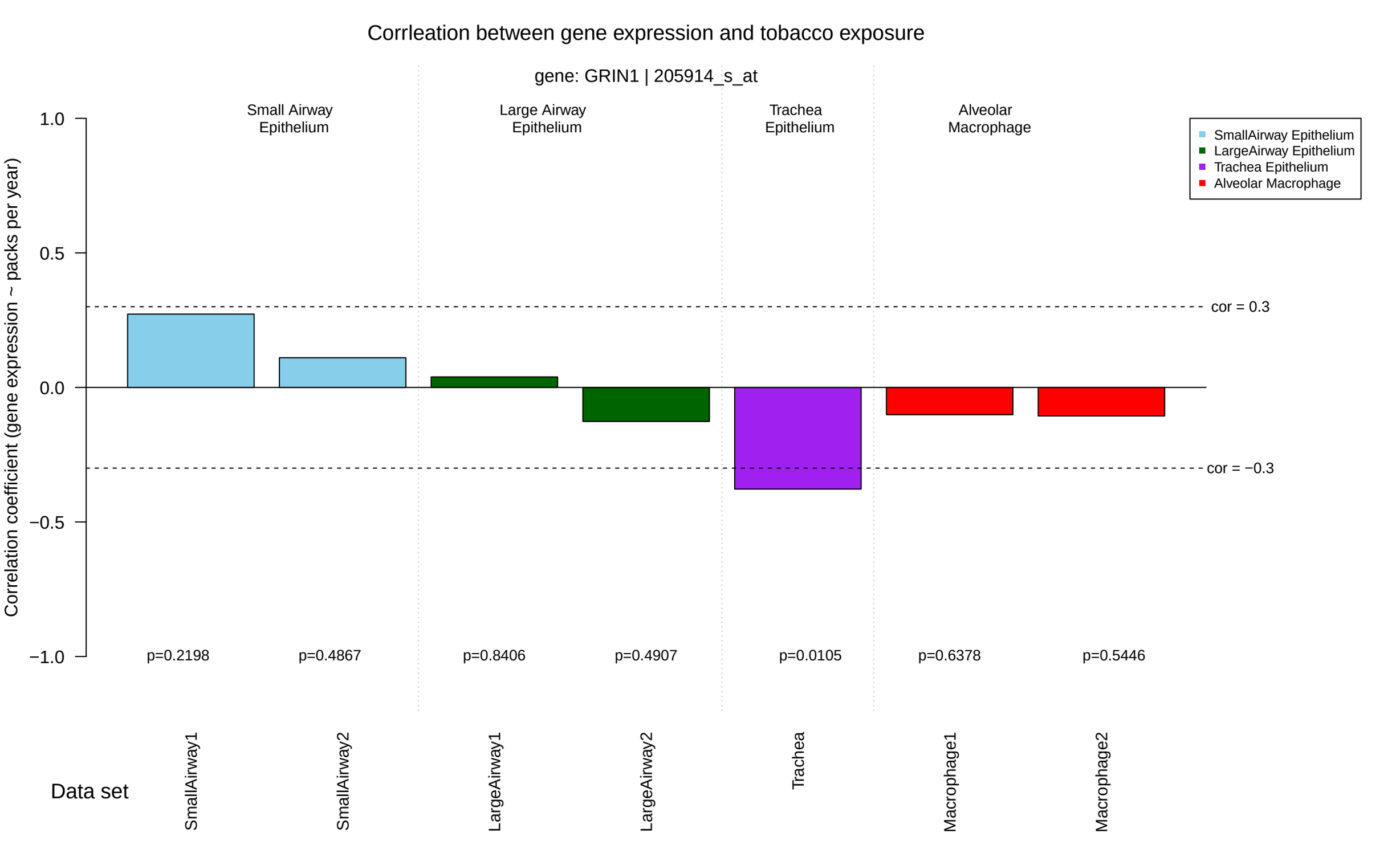

set: LargeAirway1 (p=0.1232)
GRIN1 | 205914_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6165)
GRIN1 | 205914_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0469)
GRIN1 | 205914_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8592)
GRIN1 | 205914_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7998)
GRIN1 | 205914_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIN1 | 205914_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2198
p=0.4867
p=0.8406
p=0.4907
p=0.0105
p=0.6378
p=0.5446
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2